

Module 3

Outcome-Based Teaching, Learning, and Assessment

Hands-on Training on Outcome-Based Education (OBE)

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Outcome of this Module

1. Understand constructive alignment between outcome, teaching-learning activities, and assessment.
2. Identify and select proper teaching-learning activities and assessment methods for the courses.
3. Design rubrics for assessment of outcomes.

Rationale for OBE

- Curriculum should be outcome-based and consistent with the qualifications framework for higher education (BAC).
- Defined learning outcomes state what learners will be able to do whether learned in a classroom, practical or industrial training, or less formal examination contexts.
- Accreditation expects outcome-based education and evidence of outcomes assessment processes and results (BAETE).
- OBE creates a clear link between curriculum, teaching-learning, assessment and continuous improvement.

OBE shifts emphasis from content delivery to demonstrable learner achievement.

Components of OBE

Program Educational Objectives (PEO)

Attainment in about 5 years after graduation
Long-term graduate achievements

Program Outcomes (PO)

Attainment by the time of graduation
Graduate attributes required by the program

Course Outcomes (CO)

Upon completion of a course
Measurable knowledge, skills and attitudes

PEO, PO and CO must work together as an aligned outcome system

Traditional Education Process vs OBE

Traditional Educational Process

- Input and process driven.
- Assessment mainly checks exams, quiz, and completion.
- Quality control relies largely on teaching evaluation and process monitoring.

Outcome-Based Education

- Input, process, output and stakeholder feedback are connected.
- Focuses on what students can know, do and demonstrate at graduation.
- Assessment is aligned with outcomes and improvement is evidence-based.

Focus

Inputs vs Outcomes

Design

**Teaching First vs
Outcome First**

Assessment

**Completion vs
Attainment**

Improvement

Periodic Review vs CQI

Changes Required

1. Course Content

Review content to suit specified learning outcomes, current development, industrial needs and accreditation requirements.

2. Teaching-Learning Methods

Introduce innovative/flexible methods and delivery tools to develop PEO and PO in graduates or students.

3. Assessment and Evaluation Tools

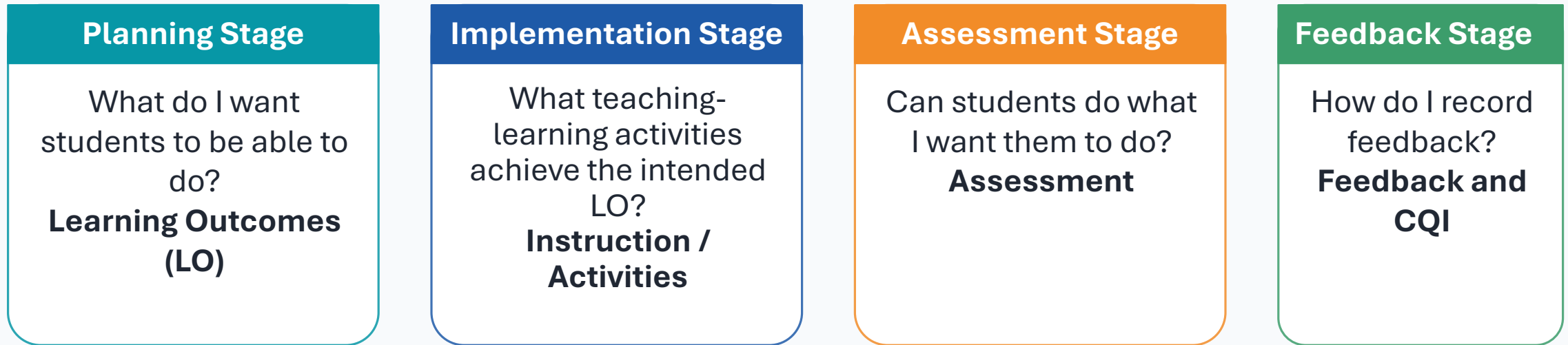
Use diverse tools to measure achievement of PEO and PO.

4. Data and Evidence Collection

Collect evidence of process and achievement.

5. Create an effective OBE management system

Constructive Alignment



Alignment means outcomes, activities and assessment support each other.

Constructive Alignment: What Can Go Wrong?

- No activities or formative assessments are provided to prepare students for the summative assessments.
- Outcome targets a performance level, but assessments or instructional strategies focus on a different level.
- Target level is high on Bloom's taxonomy, such as analysis, while activity/assessment remains at a low level, such as remember.
- A weak alignment makes evidence of attainment difficult to defend.

Different Types of Learners

Visual

- See and visualize
- Use words to evoke a visual image
- Prefer diagrams and pictures

Aural

- Natural listeners
- Prefer explanations
- Listen and verbalize

Read / Write

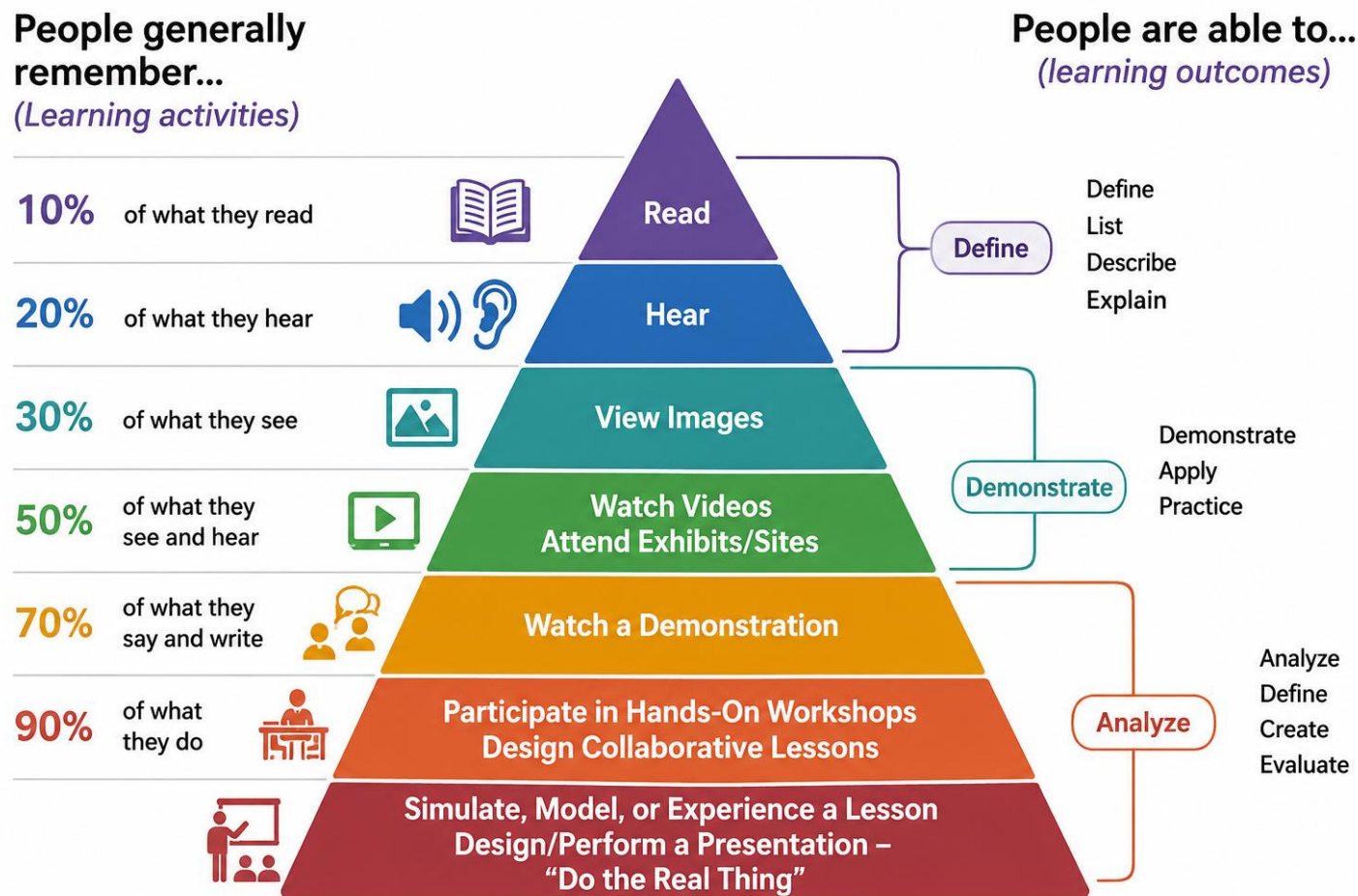
- Prefer written text
- Emphasize text-based input and output
- Enjoy reading and writing

Kinesthetic

- Use all senses to engage in learning
- Learn through trial and error
- Slow to decide; do and solve

Use a mix of activities so that learning does not depend on only one delivery mode.

The Learning Pyramid



Course Outcomes (CO)

- Course Outcomes articulate what students should be able to know, do and value by the end of a course.
- A course usually has several COs; 3 to 5 is considered a good practice.
- Learning taxonomies help construct COs strategically and measurably. Frameworks may include Bloom's, Fink's, SOLO, or discipline-specific taxonomies.
- Bloom's taxonomy is used as an example.

Learning Domains

Cognitive

- Focused on intellectual skills such as critical thinking, problem solving and creating a knowledge base.

Affective

- Focused on attitudes, values, interests and appreciation of learners.

Psychomotor

- Focused on the ability to physically accomplish tasks and perform movement and skills.

A well-written CO can address one or more learning domains, but the evidence must be measurable

Importance of Taxonomy

- Taxonomy helps select the right action verb.
- It supports alignment among CO, teaching-learning activity and assessment.
- Higher levels require richer evidence and stronger assessment methods.

Bloom's Taxonomy: Cognitive Domain

Remember	Understand	Apply	Analyze	Evaluate	Create
Retain and recall knowledge	Transfer and interpret knowledge	Use knowledge in different situations	Break information into parts	Judge value based on criteria	Generate new information or product

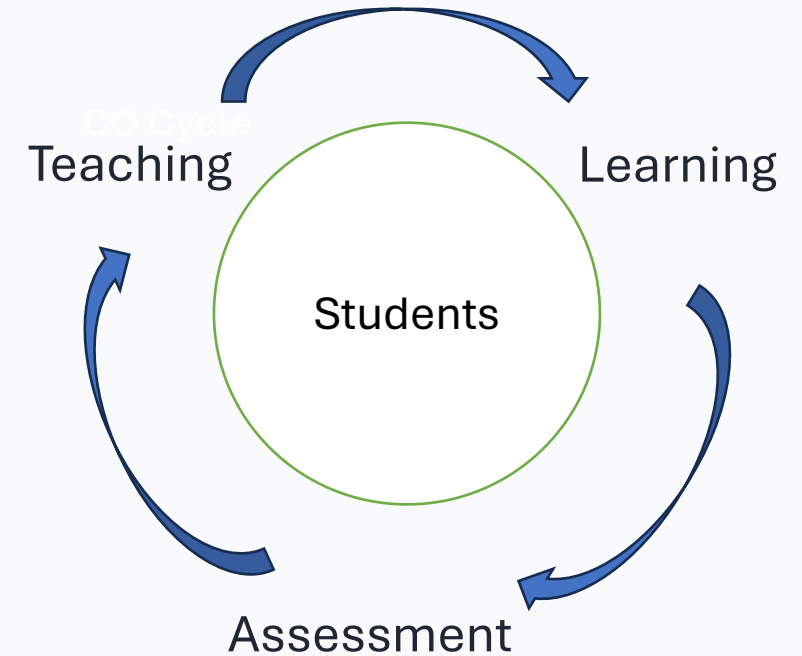
COs and assessments should indicate the desired level clearly

Cognitive Domain: Level Descriptions

Level	Meaning
Remember	Recalling relevant terminology, facts or procedures.
Understand	Grasping the meaning of information that has been presented.
Apply	Using learned information in different situations or problem solving.
Analyze	Breaking information down into parts and examining relationships.
Evaluate	Judging value based on criteria.
Create	Producing new and original thoughts, ideas, processes or products.

Course Outcomes: Wording Matters

- The wording of CO is important to consider.
- Action verbs such as **write**, **summarize** and **appraise** connect to clear learning behavior than **understand or know**.
- **Specific learning outcomes help students understand the kind of learning they need to demonstrate.**
- **The instructor should ask: What assessment or learning activity will help students reach the intended learning outcome?**
- Teaching, learning and assessment must be connected in a cycle.



How to Write COs: SMART

S	M	A	R	T
Specific Balance between too generic and overly specific.	Measurable Know whether a requirement has been fulfilled and how well.	Achievable Realistic expectation of students.	Relevant Align with program objectives or industry needs.	Time-bound Achievable within the course timeframe.

How to Write COs - Specific

Example 1

By the end of the course, a student should be able to write a program for a Graphical User Interface-based project.

Example 2

By the end of the course, a student should be able to write a Java Swing-based program with at least five different components in the content pane and handle action events like mouse clicks.

Example 3

By the end of the course, a student should be able to develop a simple GUI-based program satisfying the requirements of different components and their interactions.

How to Write COs - Measurable

- **Each CO should relate to assessment questions or activities as evidence of learning.**
- An alignment table or matrix helps determine whether all COs are assessed in a course.
- Some COs are straightforward to measure at lower cognitive levels.
- Other **COs require rubrics** because direct measurement is more complex.
- **Example concern: “Understand electrical circuits” - how will understanding be measured?**

How to Write COs - Achievable

- Select the **right level** for your students within a **12/14-week course**.
- Consider contextual factors such as **class size, course type, student background, year of the course, level of the program, number of instructors, teaching assistants and support**.
- Rethink what you can help students learn and how you can assess course learning outcomes in a **sustainable way**.

How to Write COs: Basic Structure

Action Verb	Subject Content	Level of Achievement	Condition
Select an action verb using a Suitable taxonomy	Select the subject content students perform the task for	Specify performance quality, level or expected extent	State condition of performance, if applicable

“ CO example

Design an optimized finite-state machine for a given digital control problem that satisfies the specified state transitions while minimizing the number of logic ICs used.

- Action verb:** Design
- Subject content:** Finite-state machine
- Level of achievement:** Optimized by minimizing the number of logic ICs
- Condition:** Specified state transitions

Teaching-Learning-Assessment (Remember)

Course Outcome

Recall the laws associated with probability and various discrete and continuous distributions.

Learning Activities

Lectures introducing different statistical distributions and probability axioms.

Assessment

MCQ/True-false questions addressing differences in statistical distributions and problem solving.

Teaching-Learning-Assessment (Understand)

Course Outcome

Recognize that mechanical engineering solutions need to take into account other considerations, such as societal and environmental impacts.

Learning Activities

Small presentations or speeches after a lecture, followed by student pairing and peer feedback.

Assessment

Case study report on a significant mechanical engineering milestone highlighting social and environmental impacts.

Teaching-Learning-Assessment (Apply)

Course Outcome

Apply programming fundamentals and elementary numerical methods within engineering computation.

Learning Activities

Computer lab activities with instant feedback on the code that is developed.

Assessment

Students use programming software developed at the lab, followed by in-class quizzes on using the software.

Teaching-Learning-Assessment (Analyze)

Course Outcome

Analyze the performances of different cloud architecture for large-scale application deployment.

Learning Activities

Term-long projects with multiple components deployed separately in a cloud setting.

Assessment

Report showing performance analysis through graphs and tables, including throughput, latency and compute.

Teaching-Learning-Assessment (Evaluate)

Course Outcome

Interpret data collected using hydrological field method techniques.

Learning Activities

Hands-on demonstrations of field exercises with detailed manuals on basic field techniques.

Assessment

Tutorial field exercises requiring students to collect and interpret water quality data.

Teaching-Learning-Assessment (Create)

Course Outcome

Design a site plan for a small-scale development including terrain, microclimates and infrastructure.

Learning Activities

Students visit prospective sites for their site plan project and collect pertinent data.

Assessment

Group site plan with diagrams/photos and justification for design choices.

Cognitive Domain: Action Verbs

Remember	Understand	Apply	Analyze	Evaluate	Create
Cite	Add	Acquire	Analyze	Appraise	Abstract
Define	Approximate	Adapt	Audit	Assess	Animate
Describe	Articulate	Allocate	Blueprint	Compare	Arrange
Draw	Associate	Alphabetize	Breadboard	Conclude	Assemble
Enumerate	Characterize	Apply	Break down	Contrast	Budget
Identify	Clarify	Ascertain	Characterize	Counsel	Categorize
Index	Classify	Assign	Classify	Criticize	Code
Indicate	Compare	Attain	Compare	Critique	Combine
Label	Compute	Avoid	Confirm	Defend	Compile
List	Contrast	Back up	Contrast	Determine	Compose
Match	Convert	Calculate	Correlate	Discriminate	Construct

Cognitive Domain: Learning Activities and Assessments

Level	Typical Learning Activities	Assessment Examples
Remember	Flashcards, Highlight key words list, Memory activities, Reading materials	Fill-in-the blanks, Label, Match, MCQ, Quizzes True and false questions
Understand	Case studies, Demonstrations, Diagrams, Flowcharts	One-minute paper, Presentation, Provide examples, Quizzes, Short answers
Apply	Calculate, Case studies, Creating examples, Demonstrations, Lab experiments	Discussion board post, Lab reports, One-minute paper, Presentation, Problem-solving tasks, Short answers
Analyze	Case studies, Compare and contrast, Concept map, Discussions, Group investigation	Analysis paper, Case studies, Evaluation criteria, Critique hypothesis, procedures etc.
Evaluate	Compare and contrast, Journal, Pros and cons list, Mind map, Review paper	Discussions, Presentation, Provide alternative solutions, Report
Create	Decision-making tasks, Develop new solutions or plans, Presentations, Research projects, Assignment	Develop criteria to evaluate product or solution, Outline alternative solutions, Research proposal

Bloom's Taxonomy: Affective Domain

Level	Core Meaning	Example Verbs
Receiving	Being willing to listen and receive knowledge	ask, choose, follow, listen
Responding	Actively participating and giving a response	answer, assist, discuss, perform
Valuing	Attaching value and showing motivation	accept, commit, prefer, justify
Organization	Integrating and organizing values	arrange, compare, defend, prioritize
Characterization	Acting consistently with a value system	display, influence, practice, revise

Affective Domain: Activities and Assessments

Level	Learning Activity	Assessment
Receiving	Attend presentations, listen to peers	Feedback form, one-minute paper
Responding	Participate in discussion or role play	Oral questions, group work
Valuing	Debate or defend a stance	Opinion piece, report on values
Organization	Compare value systems	Essay, presentation, group project
Characterization	Behave consistently as a team member	Peer evaluation, project behavior evidence

Teaching-Learning-Assessment: Affective Domain

Receiving

CO: Listen to other students with respect.

Assessment: Be an audience member to another student's presentation and write a summary.

Responding

CO: Speak effectively in front of an audience and actively respond to others.

Assessment: Present in class and answer peer questions.

Valuing

CO: Demonstrate and explain own values regarding environmental, societal and ethical uses of AI.

Assessment: Write an opinion piece with reasons supporting the stance.

Organization

CO: Compare value systems and understand evidence behind values in designing user interfaces for conservative and pluralistic societies.

Assessment: Organize and compare cultural value systems.

Teaching-Learning-Assessment (Characterization)

Course Outcome

Work well in a team of peers.

Assessment / Activity

A group project, including group work on an assignment.

Psychomotor Domain

Perception	Set	Guided Response	Mechanism	Complex Overt Response	Adaptation	Origination
Use of senses to obtain clues	Readiness to take action	Knowledge of the steps required to perform a task	Perform tasks in habitual manner	Skillful performance of motor acts	Skillful performance of motor acts and modification of movement in problematic or new situation	Creating new movement patterns for problematic or new situation; creates new tasks that incorporate learned ones
Choose Describe Detect Differentiate Distinguish Identify Isolate, Relate Select Separate	Begin Display Explain Move Proceed React Respond Show Start Volunteer	Assemble Build Calibrate Construct Dismantle Display Dissect Fasten Fix Grind Heat Manipulate Measure Mend Mix Organize Sketch Work	Assemble Build Calibrate Construct Dismantle Display Dissect Fasten Fix Grind Heat Manipulate Measure Mend Mix Organize Sketch Work	Assemble Build Calibrate Construct Dismantle Display Dissect Fasten Fix Grind Heat Manipulate Measure Mend Mix Organize Sketch Work	Adapt Alter Change Rearrange Reorganize Revise Vary	Arrange Combine Compose Construct Design Originate

Psychomotor Domain: Activities and Assessments

Level	Learning Activities	Assessment
Set	Observe, prepare, handle instruments	Self-criteria, oral summary
Guided Response	Copy, follow, try, practice	Evaluate accuracy against standard
Mechanism	Demonstrate and refine procedures	Performance test
Complex Overt Response	Operate or perform complex tasks	Checklist, report, practical exam
Adaptation	Modify procedures or approach	Assess outcome and self-criteria
Origination	Design or construct a new product	Assess public/private criteria

Program Outcomes - An Engineering Program

Category	POs	Related Knowledge / Attributes	
Analysis of problems and synthesis of solution	PO1 Engineering Knowledge PO2 Problem Analysis PO3 Design/Development PO4 Investigation PO5 Modern Tool Usage	WK1-WK4 WK1-WK4 WK5 WK8 WK2, WK6	Complex Engineering Problem Solving (PO1-PO6)
Responsibility	PO6 Engineer and Sustainability PO7 Ethics PO8 Individual and Teamwork	WK1, WK5, WK7 WK9 WK9	
Required in the workplace	PO9 Communication PO10 Project Management and Finance PO11 Lifelong Learning PO12 Entrepreneurship	Complex Engineering Activities (mapped with PO9)	

Complex Engineering Problem Attributes

Code	Attribute
WP1	Depth of knowledge required
WP2	Range of conflicting requirements
WP3	Depth of analysis required
WP4	Familiarity of issues
WP5	Extent of applicable codes
WP6	Extent of stakeholder involvement and needs
WP7	Interdependence

WP1 + Some (at least 2) or All from WP2-WP7

Complex Engineering Activities

Complex activities mean (engineering) activities or projects that have some or all of the following characteristics:

Code	Attribute
EA1	Range of resources
EA2	Level of interaction
EA3	Innovation
EA4	Consequences to society and the environment
EA5	Familiarity

These attributes may be encountered during capstone design, final year project, laboratory project or industry experience.

Example of CO-PO Mapping

Embedded Controller Technology

CO1: Illustrate the architecture of the microcontroller. **PO1 (C)**

CO2: Interpret the M68HC11 instruction sets. **PO2 (C)**

CO3: Design and develop firmware using assembly language. **PO3 (C)**

CO4: Design a basic hardware based on 68HC11 microcontroller. **PO5 (P)**

CO5: Work in a team and communicate effectively. **PO9 (A)**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	C/X											
CO2		C/X										
CO3			C/X									
CO4					P/X							
CO5									A/X			

CO Assessment in Courses

- **CO assessment and assessment in the course (grades) are different.**
- Decide which courses should be considered for assessment.
- Assessment may cover all courses.
 - It may be for a selective section in case of multiple sections or a selective semester/trimester.
- Even when selective core courses are assessed, all courses still require course-level CQI.

CO Assessment in Theory Courses

- Class tests/quizzes may be excluded
 - if all CTs are not counted (e.g., 3 out of 4).
 - Midterm/final always covers the whole syllabus.
 - No need to include the same question/content twice.
 - **Too many assessments make life difficult.**
 - **Assignments are useful if open-ended and individual problem.**
- First, we need to prepare a course syllabus that must have
 - COs
 - CO with assessment methods
 - CO-PO mapping

CO Assessment - Algorithms

CO	Course Outcome	PO
CO1	Analyze running times of algorithms using asymptotic analysis.	PO1 (C)
CO2	Identify different algorithm paradigms and analyze performance.	PO2 (C)
CO3	Compare data structures and select an appropriate structure.	PO2 (C)
CO4	Explain complexity class and relationship among them.	PO1 (C)

CO	Assessment Method	%
-	Attendance	05
-	Assignment	05
-	Class Tests	20
CO1, CO2, CO3	Mid Exam	30
CO2, CO3, CO4	Final Exam	40

Mapping of Questions to COs

- All assessments (CT, assignments, midterm or final) considered for attainment must map to COs.
- For mid/final examinations, mapping can be included with the question paper or question moderation form.
- Question-to-CO mapping clearly indicates whether the question paper is balanced.
- Mapping also helps determine which COs are strongly or weakly assessed.

Question Moderation Form

CO	Course Outcome	PO
CO1	Analyze running times of algorithms using asymptotic analysis.	PO1 (C)
CO2	Identify different algorithm paradigms and analyze performance.	PO2 (C)
CO3	Compare data structures and select an appropriate structure.	PO2 (C)
CO4	Explain complexity class and relationship among them.	PO1 (C)

Q.	CO	Remember	Understand	Apply	Analyze	Evaluate	Create
Q1(a)							
Q1(b)							
Q2							
Q3							
Total							

Use the form to check CO coverage and Bloom's level allocation before the final question paper is approved.

Question Evaluation

- 1. Relevance of questions according to the levels of cognitive domain in Bloom's taxonomy.**
- 2. Relevance of the learning outcomes to the question provided.**
- 3. Breadth of course material expected to be covered during the semester.**
4. Clarity of the question provided.
5. Distribution of marks allotted for each question.
6. Correctness of the grammar and spelling.
7. Format followed as prescribed by the program.

Mapping of Questions to COs: Good Practice

- Keep the work as simple as possible.
- **All questions may not be mapped. (Matter of Dispute)**
- **The question we map (for Attainment) must be compulsory.**
- **All optional questions must have same CO(s).**
- **One question should not map multiple COs.**
- **Avoid too many breakdowns of mapping.**

CO Assessment - Lab Courses

- Assessment is often not as simple as theory courses.
- **Evaluation components may include lab final, lab report, quiz, performance, viva, project/design report and presentation.**
- **No need to bring everything in the CO assessment.**
- CO assessment and course assessment/grades are different.

CO Assessment - Lab Courses Example

PO	Assessment
PO4: Investigation	Open-ended projects
PO5: Modern Tool Usage	Tools taught in lab classes
PO8: Individual and Teamwork	Group projects
PO9: Communication	Viva, project presentation, report writing

CO Assessment - Digital Electronics Lab

CO	Course Outcome	PO
CO1	Implement combinational circuits using hardware tools.	PO5 (P)
CO2	Implement sequential circuits using flip-flops and counters.	PO5 (P)
CO3	Work in a team and communicate effectively.	PO8 (A)

CO	Assessment Method	%
CO1, CO2	Lab performance	35
CO3	Simulation/design task	25
CO4	Presentation/report	15
All	Viva/quiz	25

Final Year Design Project

- **Final Year Design Project or the Capstone project is a key element in outcome-based teaching and learning.**
 - Extending over a period of one year.
 - Represents a culminating demonstration of the program outcomes at the level of solving complex engineering problems.
 - Based on the knowledge and skills acquired in earlier coursework.
 - Should incorporate appropriate engineering standards and multiple realistic constraints.
- **The Final Year Design Project should address most of the POs.** In fact, this is the primary means to show achievements of a number of PO's, e.g., Investigation (PO 4), Engineer and Sustainability (PO 6), Individual and Teamwork (PO 8), Project Management (PO 10), Lifelong Learning (PO 11), Entrepreneurship (PO 12).

Final Year Design Project

- Learning Activities in the Final Year Design Project should be appropriately planned so that the Project mimics a real-life professional engineering task.
- Each activity should be geared towards achieving one or more POs.
- All the activities should be adequately documented.
- It is desirable to involve the industry in the project. Either through assigning a real-life practical design problem or through involving experts from the industry in co-supervision and/or evaluation.
- Thesis/Projects typically done by 4th-year students in our country do not satisfy the requirements of the Capstone Project.

Assessment Rubrics

- Relatively straightforward for theory courses through written exams.
- **Challenging to assess the following POs, as it is not possible to assess them through written exams:**
 - **PO 5: Modern Tool Usage**
 - **PO 8: Individual and Team Work**
 - **PO 9: Communication**
 - **PO 11: Lifelong Learning**
- **Required to include lab courses, especially the assigned projects.**
- **The rubrics are mainly useful for labs or for theory courses that have a project/presentation.**
- It will tell the students **“What We Want”** from them in a presentation, project, or report writing.

Assessment Rubrics: Designing Criteria

A **rubric** is an assessment tool that **clearly indicates achievement criteria** across all the components of any kind of student work, from written to oral to visual. It can be used for marking assignments, class participation, or overall grades.

- You find yourself rewriting the same comments on several different students' assignments.
- Your marking load is high, and writing out comments takes up a lot of your time.
- Students repeatedly question you about the assignment requirements, even after you've handed back the marked assignment.
- You want to address the specific components of your marking scheme for student and instructor use, both before and after assignment submission.
- You find yourself wondering if you are grading or commenting equitably at the beginning, middle, and end of a grading session.
- You have a team of graders and wish to ensure validity and inter-rater reliability.

How to Make a Rubric

- ✓ Decide what criteria or essential elements must be present in the student’s work to ensure that it is high in quality (exemplary student work).
- ✓ Decide how many levels of achievement you will include on the rubric and how they will relate to your institution’s definition of grades as well as your own grading scheme.
- ✓ For each criterion, component, or essential element of quality, describe in detail what the performance at each achievement level looks like.

Performance Indicator	Unacceptable 0-5	Acceptable 5.5-7	Adequate 7.5-8.5	Excellent 9-10

Rubric Example: Teamwork

Marks →	4	3	2	1	0
Contributes to Team Goals	Student provides at least one concrete example in which he/she helps the team move forward by: Offering alternative ideas which build on the ideas of others and Articulating the merits of those ideas.	Student provides at least one concrete example in which he/she helps the team move forward by Offering alternative ideas which build on the ideas of others.	Student provides at least one concrete example in which he/she offered suggestions or ideas to advance the work of the group.	Student provides at least one concrete example in which he/she offered a suggestion but it does not appear the suggestion would advance the work of the group.	Student does not appear to engage with other members of the team or does not provide examples in which he/she shared ideas to advance the work of the group.

Rubric Example: Presentation

Marks →	4	3	2	1
Oral	Speaker is fluent and poised; uses language comfortably and appropriately; speaks at an effective rate and volume; few fillers	Some degree of nervousness apparent; minor problems with language usage; speaker may speak too slowly or quickly, too loudly or softly, fillers are noticeable	Speaker seems uncomfortable; several problems with language usage; speaker speaking much too slowly or quickly, too loudly or softly, fillers are noticeable	Speaker is unable to deliver presentation coherently
Non-verbal	The speaker uses gestures comfortably in line with his/her own style; eye contact is appropriate for audience; use of space appropriate for the situation	Speaker gesturing too much or too little; eye contact may be slightly too much or too little; speaker may be moving around a little too much or not quite enough	Speaker gesturing too much or too little; using distracting gestures (e.g., playing with a ring); not enough eye contact; inappropriate use of space	Nonverbal components of the presentation distract from ability of the audience to receive the message
Visual Aids	Appropriate visual aids are used; visual aids serve as a complement to the speaker and the message to be delivered, designed effectively; speaker uses visual aid easily	Appropriate visual aids are used; a few weaknesses in design; a few difficulties with use	Choice of visual aid is poor; weaknesses with design; difficulties with use	Inappropriate choice of visual aids; design detracts from speaker's ability to deliver the message; inability of speaker to use visual aid

Rubric Example: Modern Tool Usage

Marks →	1	2	3	4
Selects and uses tools	Cannot select the appropriate tool nor understand the rationale for using a given tool Cannot use the modern engineering tool(s), even with guidance. Safety concerns may become apparent when in possession of the tool	Explains the reasoning behind the use of a particular tool, when there is only one Uses the modern engineering tool(s) given to them to complete the task with significant guidance	Justifies the selection of one tool over another, given op-tions Uses modern engineering tools that may be appropriate for the problem at hand with some guid-ance Identifies and investigates different tools for the problem	Selects the most appropriate tool for the problem Uses mod-ern engineering tools that are most appli-cable to the problem at hand, without ex-ternal guidance
Evaluates tools	Is unaware of tool's error and limitations Does not understand the role of a tool in the broader subject	Is aware of limitation of tools but does not account for them	Identifies the limitation of tools accounts for some of the limitations of the tools	Critiques these tools and justify the use of a particular one Critically evaluates and account for the limitations of tools
Creates and adapts tools, Integrates the tool range	Knows only what is told to them; no knowledge of the evolution/adaptation of the tools for increasingly complex problems	Understands that other tools exist, but cannot integrate them into a cohesive protocol	Understands that tools are often used together to expand the process, and shows adequate knowledge about integrating tools to perform a task	Expands upon tool selection, to be able to use several tools, either in series or parallel, to perform a desired function

Use of Rubrics

- **Develop a different rubric for each assignment where necessary.**
- Be transparent.
- Be prepared to revise your rubrics.
- **You may develop rubrics interactively with your students.**
- Rubrics for Assessment

Report	Individual/Group Project
Viva	Project Report
Presentation	Weekly Assignment
Prototype	Logbook

PO Assessment

CEP is already embedded in the definition of some POs

PO1

Apply knowledge of mathematics, natural science, computing, engineering fundamentals, and an engineering specialization as specified in **WK1 to WK4**, respectively, **to develop solutions to complex engineering problems.**

- Describe how the attainment of PO1 is assessed and evaluated.
- State and justify the methods, tools, criteria, and scale used in the assessment process.
- State the expected level of attainment.
- State how Knowledge Profile attributes (WK1–WK4) are incorporated in PO1. Identify which attributes of the Range of Complex Engineering Problems (WP1–WP7) are addressed through the attainment of PO1 and provide evidence to support the assertion.
- Present a summary of the results obtained after the assessment and analysis to demonstrate the extent to which PO1 is being attained.

PO Assessment

CEP is not embedded in the definition of some POs

PO 8

Function effectively as an individual, and as a member or leader in diverse and inclusive teams and in multi-disciplinary, face-to-face, remote, and distributed settings (WK9).

- Describe how the attainment of PO8 is assessed and evaluated.
- State and justify the methods, tools, criteria, and scale used in the assessment process.
- State the expected level of attainment.
- State how Knowledge Profile attribute WK9 is incorporated in PO8.
- Present a summary of the results obtained after the assessment and analysis to demonstrate the extent to which PO8 is being attained.

PO Assessment

CEA is already embedded in the definition of some POs

PO 9

Communicate effectively and inclusively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, taking into account cultural, language, and learning differences.

- Describe how the attainment of PO9 is assessed and evaluated.
- State and justify the methods, tools, criteria and scale used in the assessment process.
- Identify the expected level of attainment.
- Identify which attributes of the Range of Complex Engineering Activities (EA1–EA5) are addressed through the attainment of PO9 and provide evidence to support the assertion.
- Present a summary of the results obtained after the assessment and analysis to demonstrate the extent to which PO9 is being attained.

End of Module

Module 3: Outcome-Based Teaching, Learning, and Assessment



Thank You